

16TH Experiment

```
import cv2

import numpy as np

import matplotlib.pyplot as plt

from google.colab import files

# Upload image

uploaded = files.upload()

image_path = list(uploaded.keys())[0]

img = cv2.imread(image_path)

# Convert to RGB for plotting

img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# Grayscale

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Thresholding

_, thresh = cv2.threshold(gray, 127, 255, cv2.THRESH_BINARY)

# Kernel

kernel = np.ones((5,5), np.uint8)

# Morphological Operations

erosion = cv2.erode(thresh, kernel, iterations=1)

dilation = cv2.dilate(thresh, kernel, iterations=1)

opening = cv2.morphologyEx(thresh, cv2.MORPH_OPEN, kernel)

closing = cv2.morphologyEx(thresh, cv2.MORPH_CLOSE, kernel)

# Display all outputs in one figure
```

```
plt.figure(figsize=(12,8))
```

```
plt.subplot(2,3,1)
```

```
plt.imshow(img_rgb)
```

```
plt.title("Original")
```

```
plt.axis("off")
```

```
plt.subplot(2,3,2)
```

```
plt.imshow(thresh, cmap="gray")
```

```
plt.title("Threshold")
```

```
plt.axis("off")
```

```
plt.subplot(2,3,3)
```

```
plt.imshow(erosion, cmap="gray")
```

```
plt.title("Erosion")
```

```
plt.axis("off")
```

```
plt.subplot(2,3,4)
```

```
plt.imshow(dilation, cmap="gray")
```

```
plt.title("Dilation")
```

```
plt.axis("off")
```

```
plt.subplot(2,3,5)
```

```
plt.imshow(opening, cmap="gray")
```

```
plt.title("Opening")
```

```
plt.axis("off")
```

```
plt.subplot(2,3,6)
```

```
plt.imshow(closing, cmap="gray")
```

```
plt.title("Closing")
```

```
plt.axis("off")
```

```
plt.tight_layout()
```

```
plt.show()
```

[Choose Files](#) image_0.jpg
image_0.jpg(image/jpeg) - 2791601 bytes, last modified: 3/25/2024 - 100% done
Saving image_0.jpg to image_0 (2).jpg

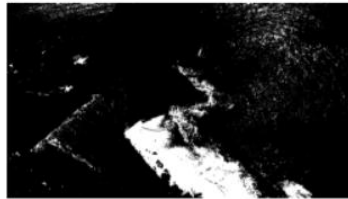
Original



Threshold



Erosion



Dilation



Opening



Closing

